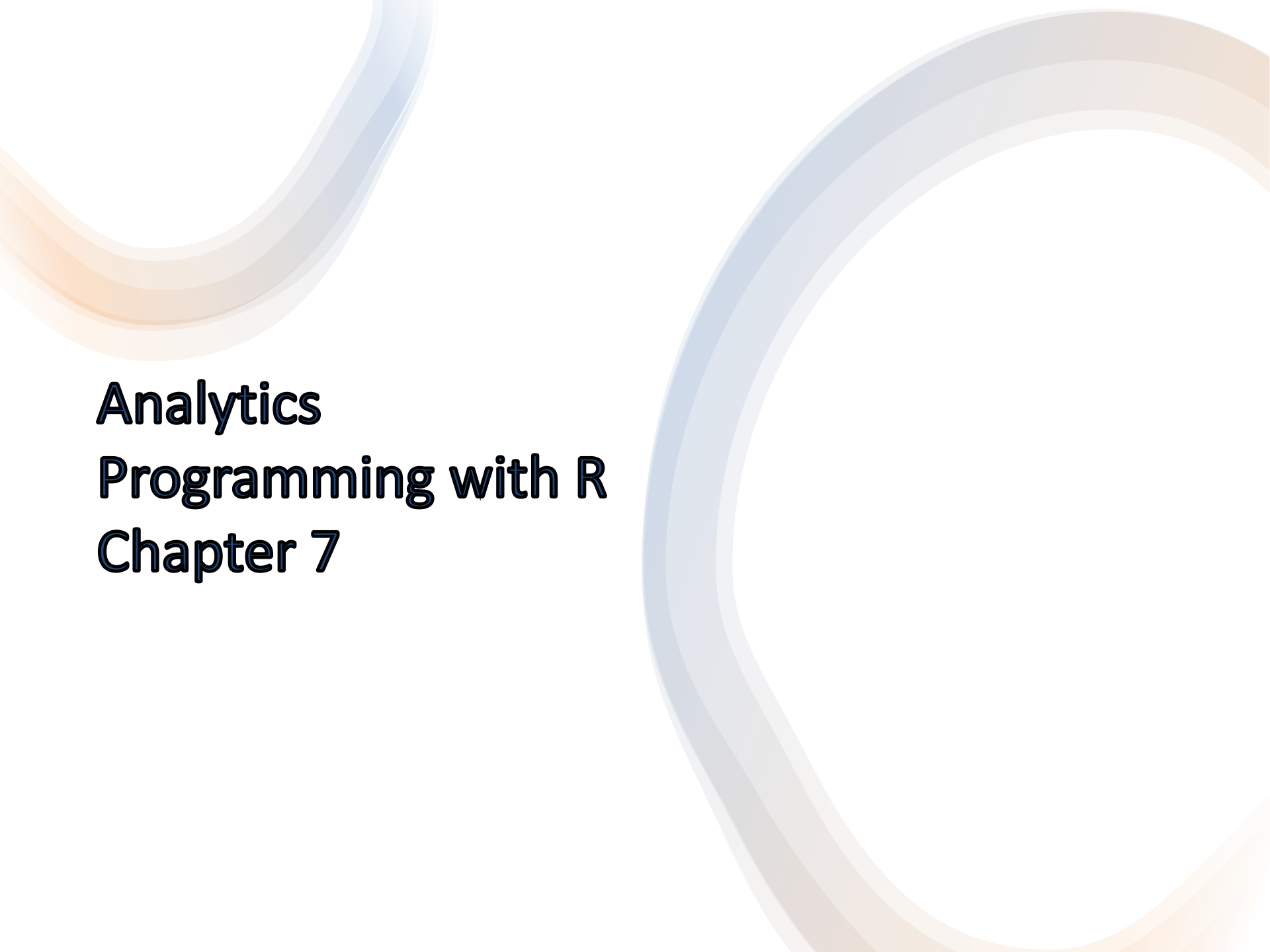




Analytics

Programming with R

Chapter 7

Data Visualization

- ggplot2
- palmerpenguins (includes penguins dataset containing body measurements for penguins on three islands in the Palmer Archipelago)
- Ggthemes (colorblind safe color palette)

```
library\(palmerpenguins\)
```

```
library\(ggthemes\)
```

```
View(penguins)
```

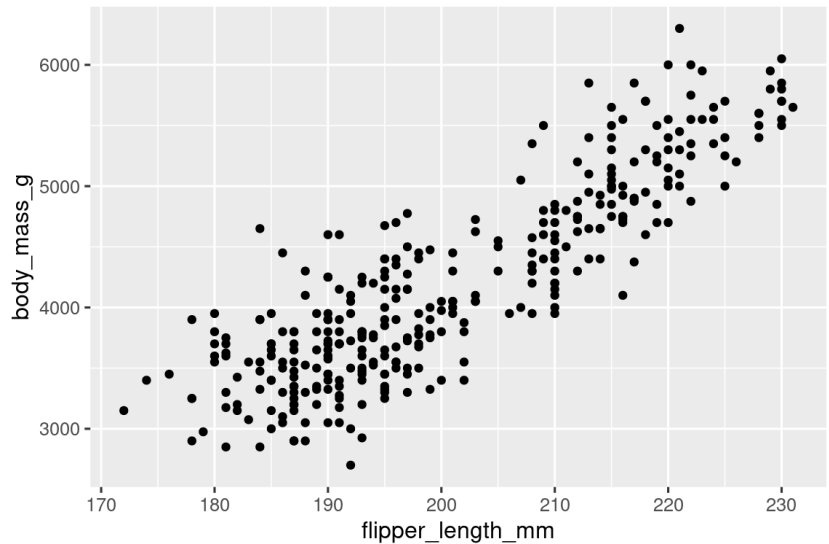


Data Visualization

- `ggplot( data = penguins, mapping = aes(x = flipper_length_mm, y = body_mass_g) ) + geom\_point()`
- `#> Warning: Removed 2 rows containing missing values (`geom_point()`).`
- The first argument of `ggplot()` is the dataset to use in the graph and so `ggplot(data = penguins)`
- The mapping argument is always defined in the `aes()` function, and the x and y arguments of `aes()` specify which variables to map to the x and y axes.
- The function `geom_point()` adds a layer of points to your plot, which creates a scatterplot
- bar charts use bar geoms (`geom_bar()`), line charts use line geoms (`geom_line()`), boxplots use boxplot geoms (`geom_boxplot()`), scatterplots use point geoms (`geom_point()`)

Data Visualization

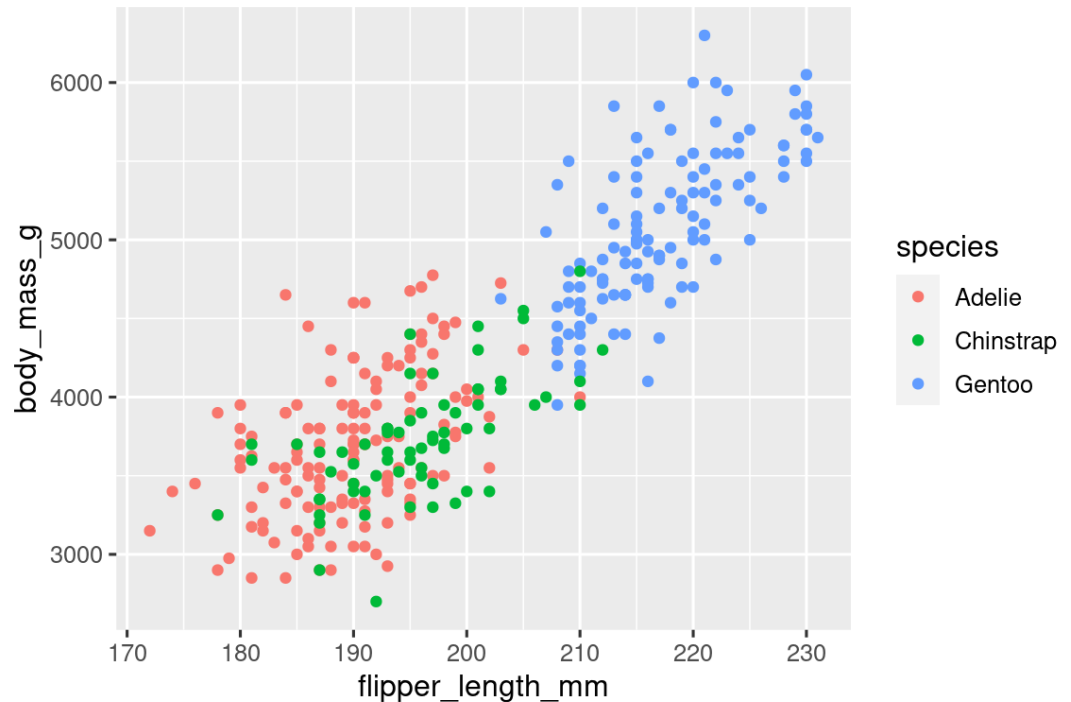
```
ggplot( data = penguins, mapping  
= aes(x = flipper_length_mm, y =  
body_mass_g) ) + geom_point()  
#> Warning: Removed 2 rows  
containing missing values  
(`geom_point()`).
```



Data Visualization

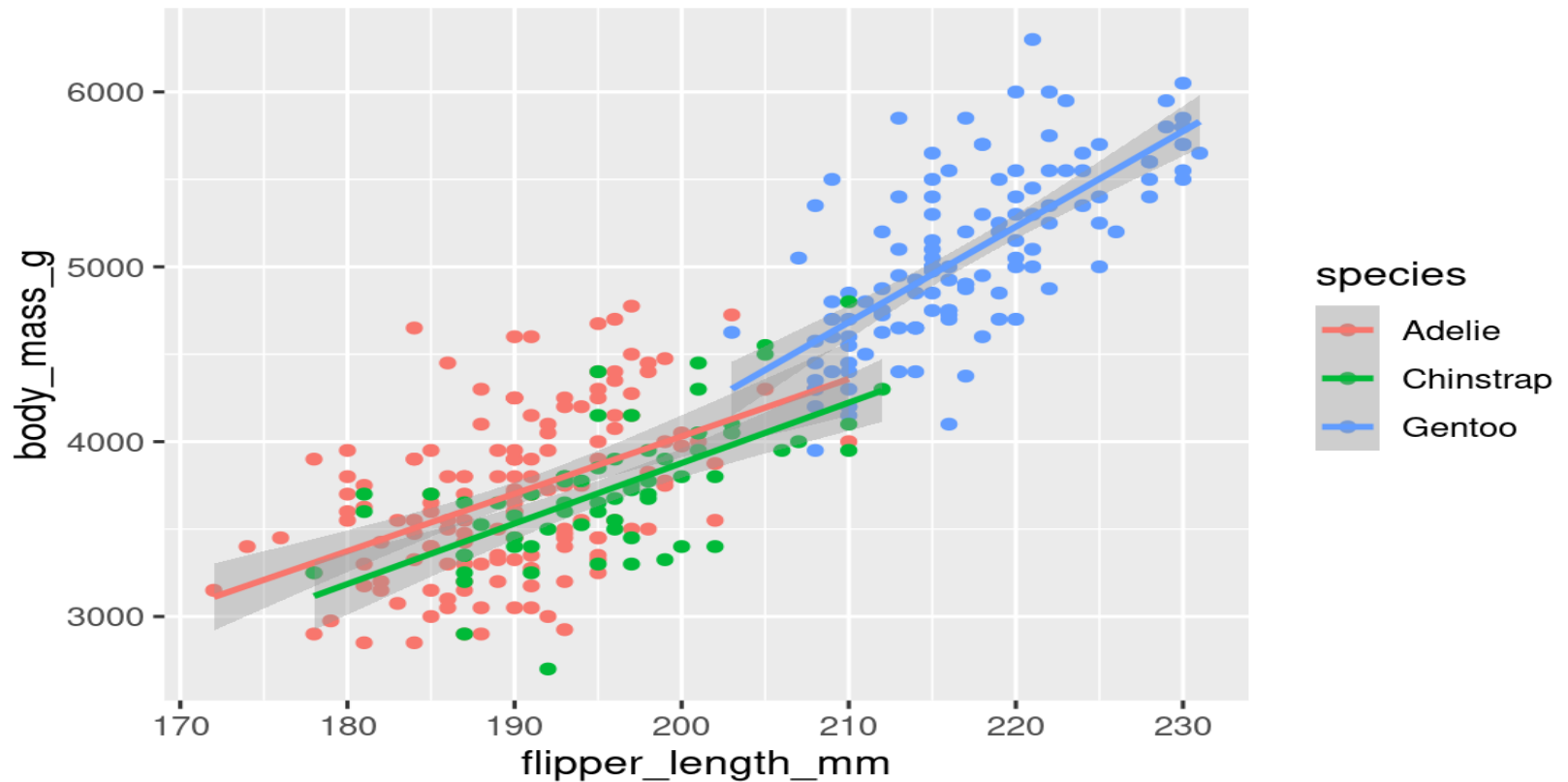
```
ggplot( data =  
penguins, mapping =  
aes(x =  
flipper_length_mm, y  
= body_mass_g,  
color=species) ) +  
geom_point()
```

```
#> Warning:  
Removed 2 rows  
containing missing  
values  
(`geom_point()`).
```



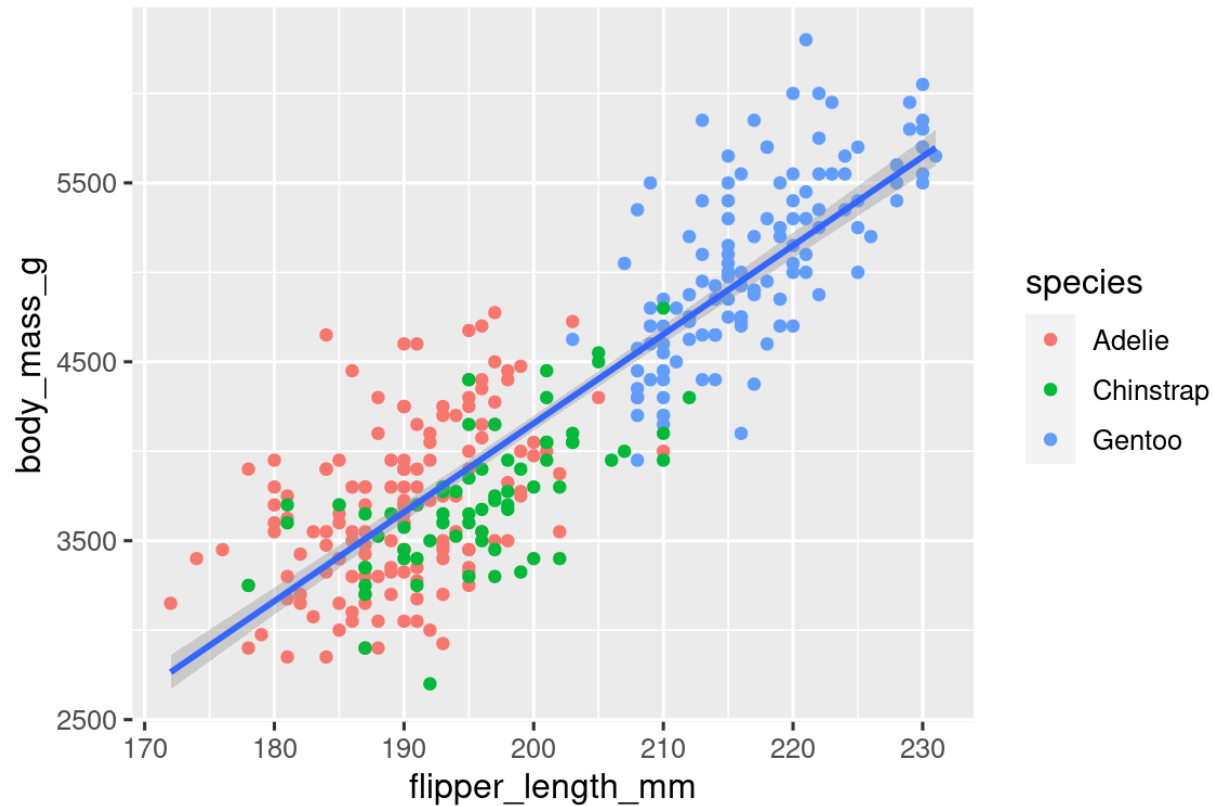
Data Visualization

```
ggplot( data = penguins, mapping = aes(x =  
flipper_length_mm, y = body_mass_g,  
color=species) ) + geom_point() +  
geom_smooth(method = "lm")
```



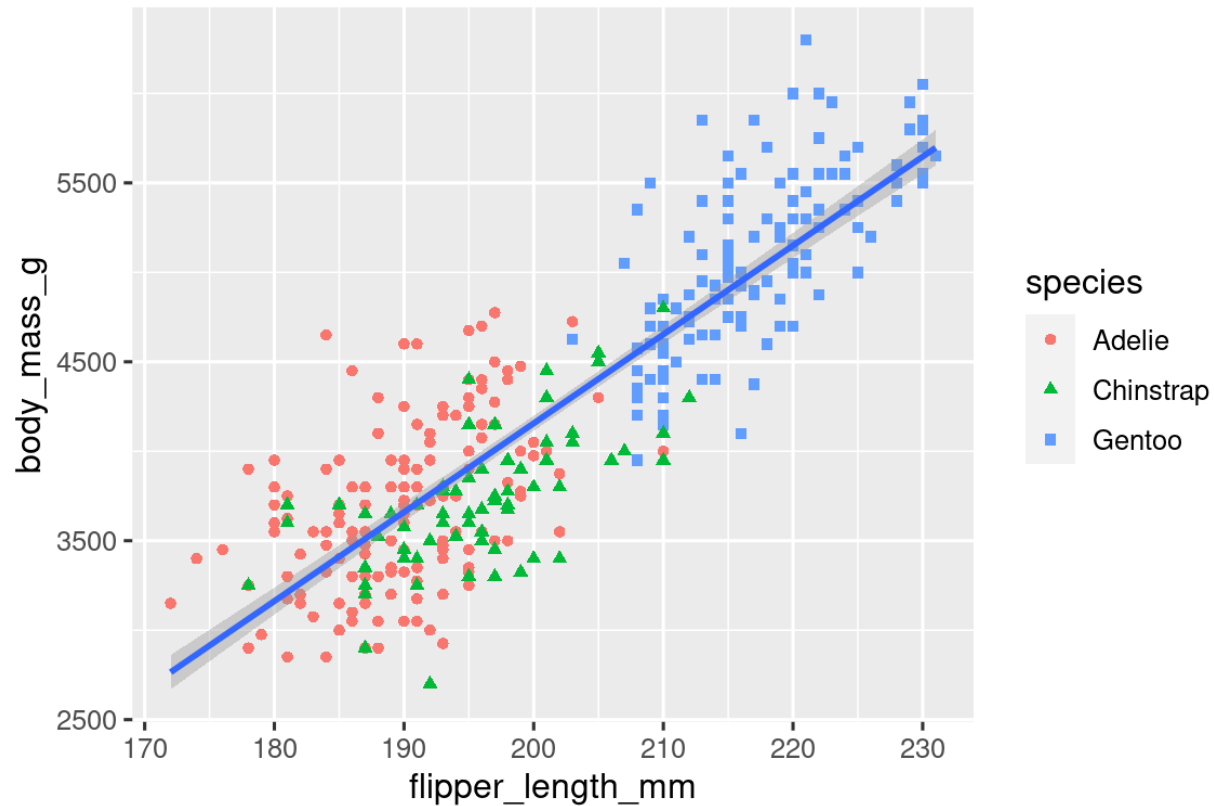
Data Visualization

```
ggplot(  
  data = penguins,  
  mapping = aes(x = flipper_length_mm, y = body_mass_g)  
) +  
  geom_point(mapping = aes(color = species)) +  
  geom_smooth(method = "lm")
```



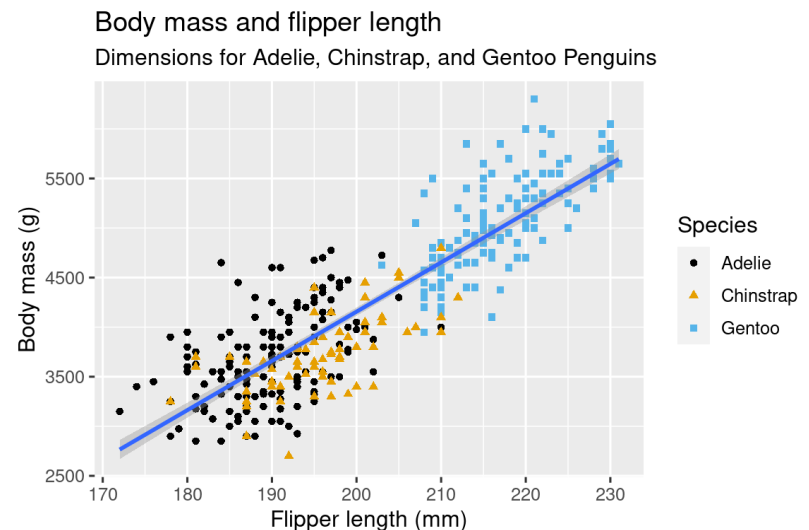
Data Visualization

```
ggplot(  
  data = penguins,  
  mapping = aes(x = flipper_length_mm, y = body_mass_g)  
) +  
  geom_point(mapping = aes(color = species, shape = species)) +  
  geom_smooth(method = "lm")
```



Data Visualization

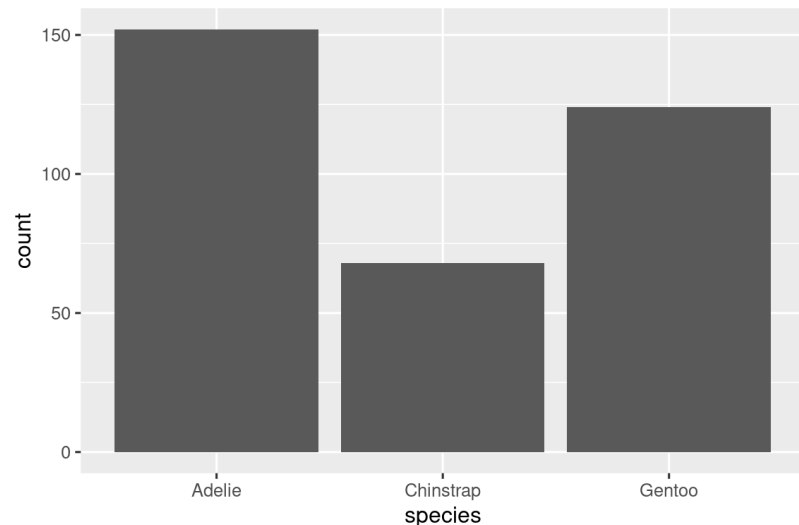
```
ggplot( data = penguins,  
  mapping = aes(x = flipper_length_mm,  
    y = body_mass_g)) +  
  geom_point(aes(color = species, shape = species)) +  
  geom_smooth(method = "lm") +  
  labs( title = "Body mass and flipper  
length",  
    subtitle = "Dimensions for Adelie,  
Chinstrap, and Gentoo Penguins",  
    x = "Flipper length (mm)", y = "Body  
mass (g)",  
    color = "Species", shape = "Species"  
  ) +  
  scale_color_colorblind()
```



Data Visualization

Distributions – Categorical Variable

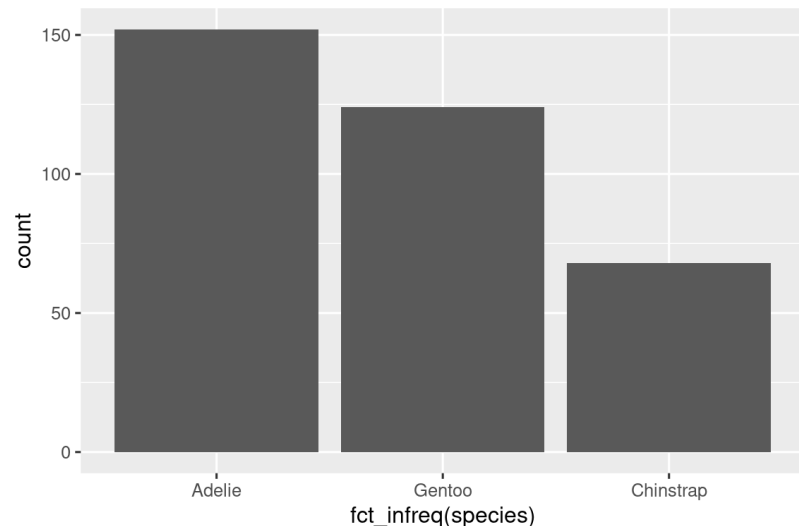
```
ggplot(penguins, aes(x =  
species)) +  
  geom_bar()
```



Data Visualization

Distributions – Categorical Variable

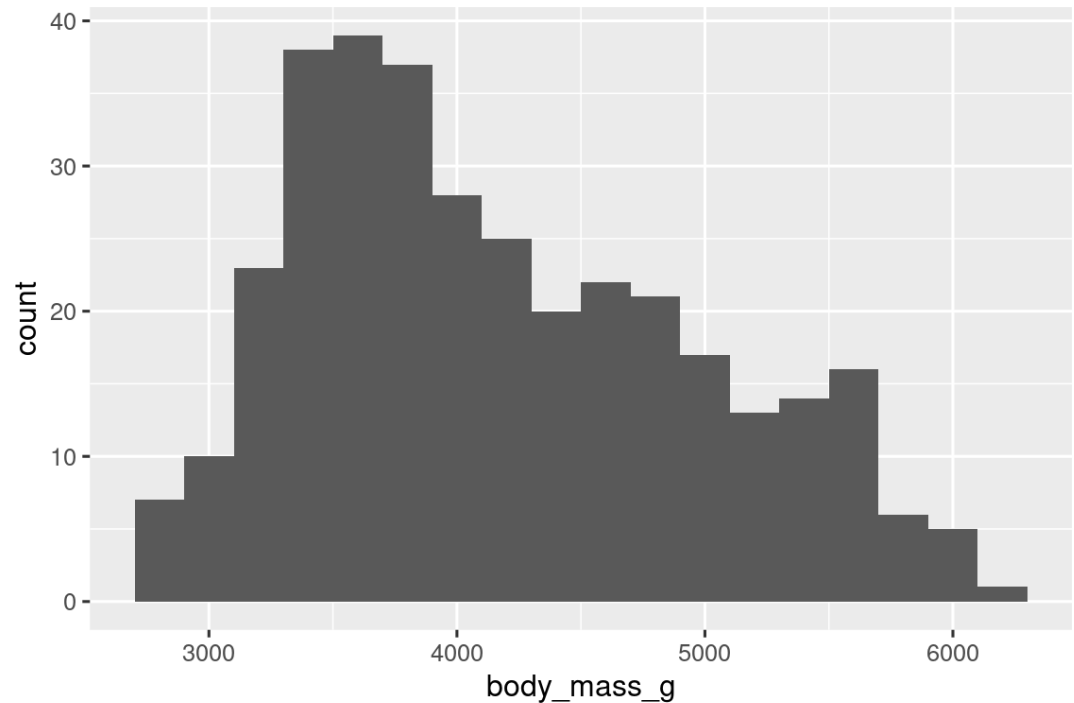
```
ggplot(penguins, aes(x =  
fct_infreq(species))) +  
geom_bar()
```



Data Visualization Distributions – Quantitative Variable

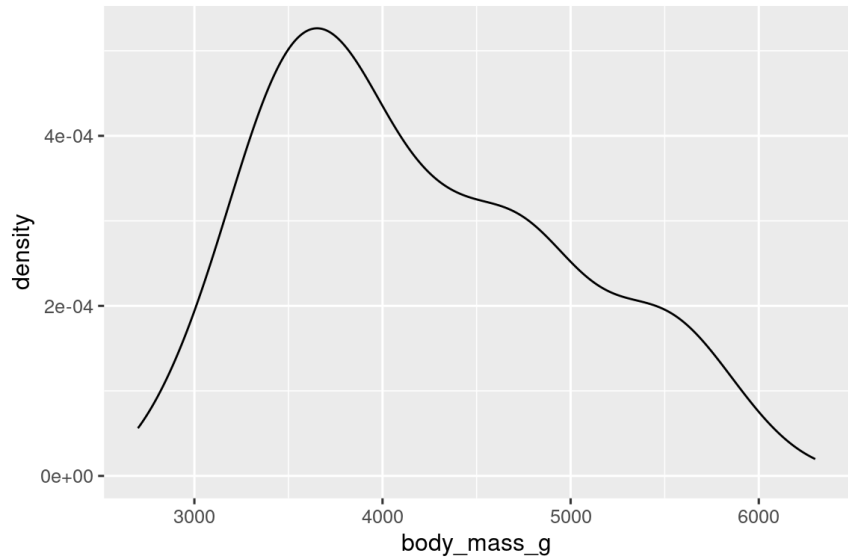


```
ggplot(penguins,  
aes(x =  
body_mass_g)) +  
geom_histogram(binwidth = 200)
```



Data Visualization Distributions – Quantitative Variable

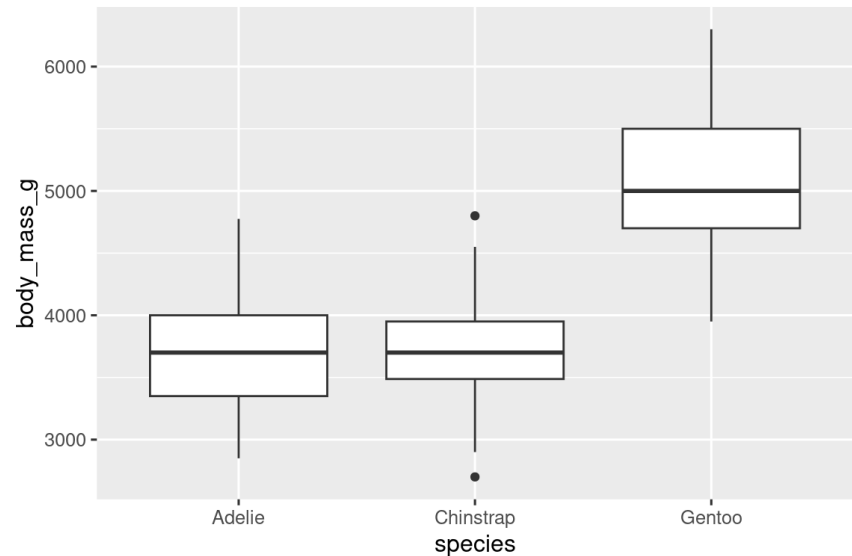
```
ggplot(penguins, aes(x =  
body_mass_g)) +  
  geom_density()
```



Data Visualization

Relationship – Quantitative and Categorical Variable

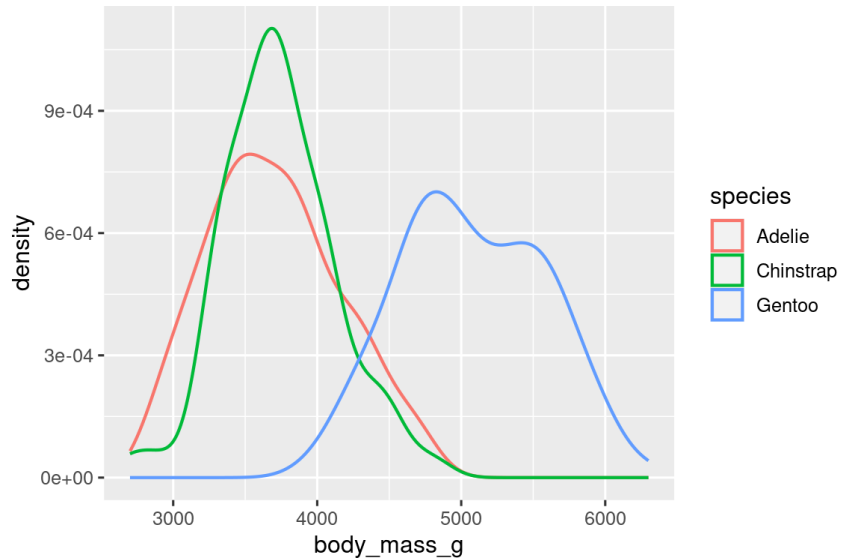
```
ggplot(penguins, aes(x =  
species, y = body_mass_g)) +  
geom_boxplot()
```



Data Visualization

Relationship – Quantitative and Categorical Variable

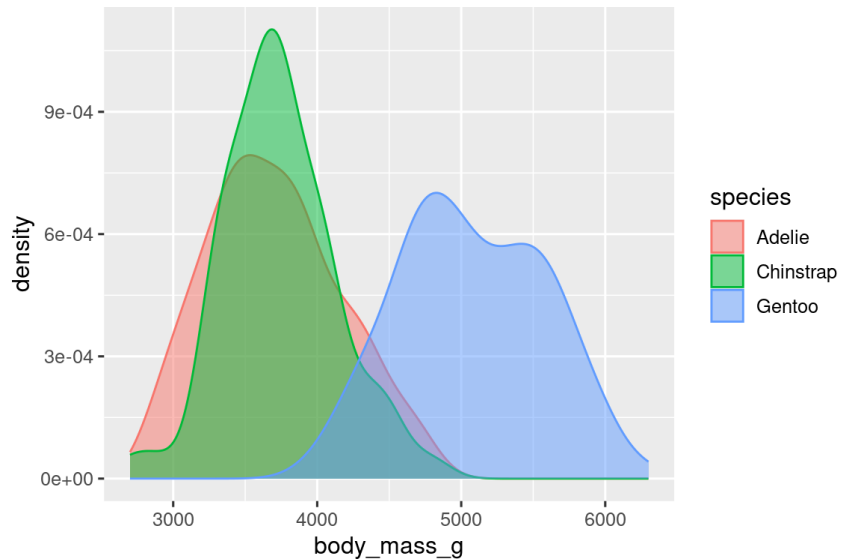
```
ggplot(penguins, aes(x =  
body_mass_g, color = species))  
+  
geom_density(linewidth =  
0.75)
```



Data Visualization

Relationship – Quantitative and Categorical Variable

```
ggplot(penguins, aes(x =  
body_mass_g, color = species,  
fill = species)) +  
geom_density(alpha = 0.5)
```

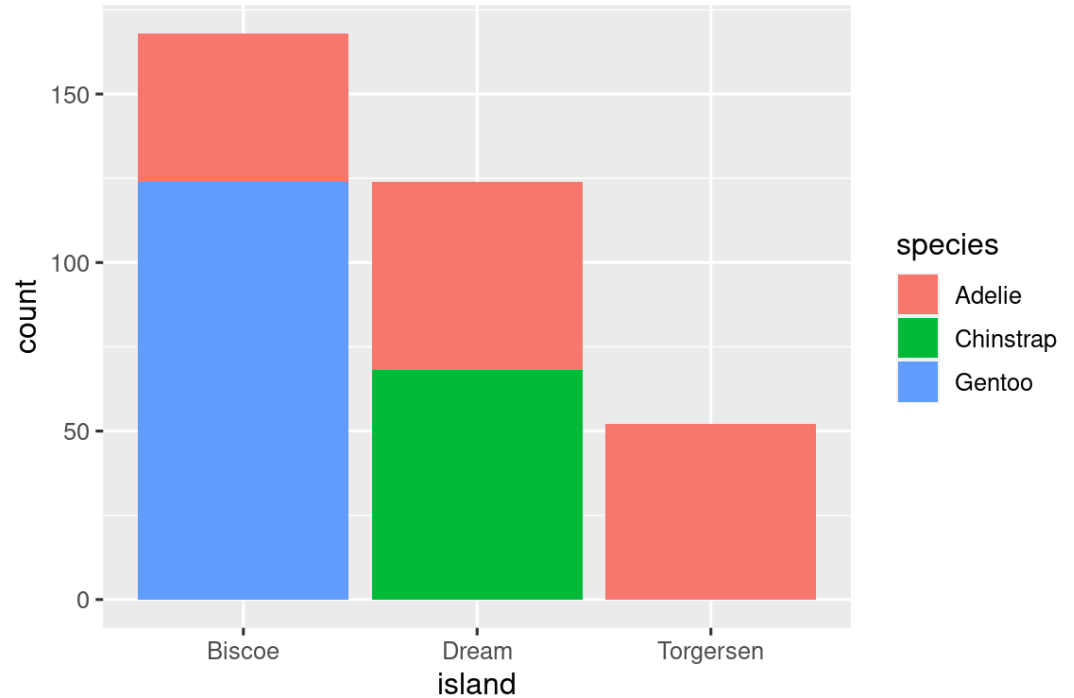


Data Visualization

Relationship – Two Categorical Variable



```
ggplot(penguins, aes(x =  
island, fill = species)) +  
geom_bar()
```

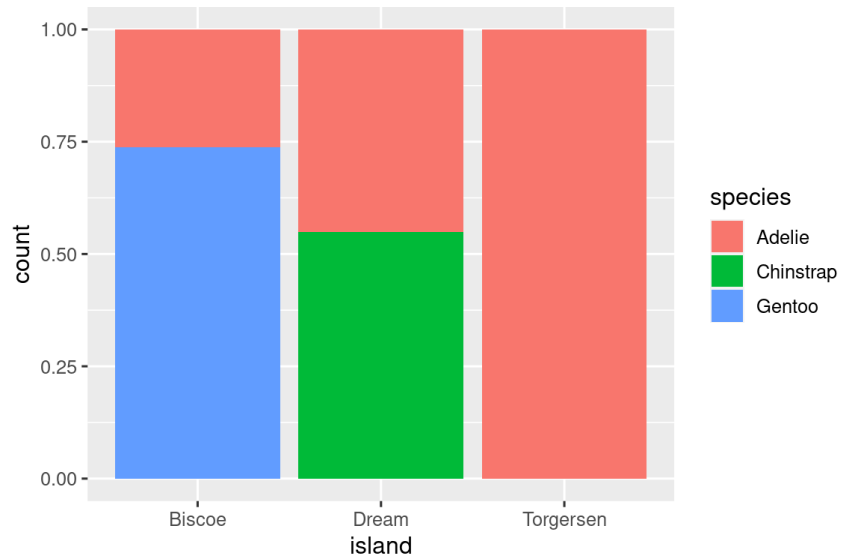


Data Visualization

Relationship – Two Categorical Variable (Relative Frequency)



```
ggplot(penguins, aes(x = island,  
fill = species)) +  
geom_bar(position = "fill")
```

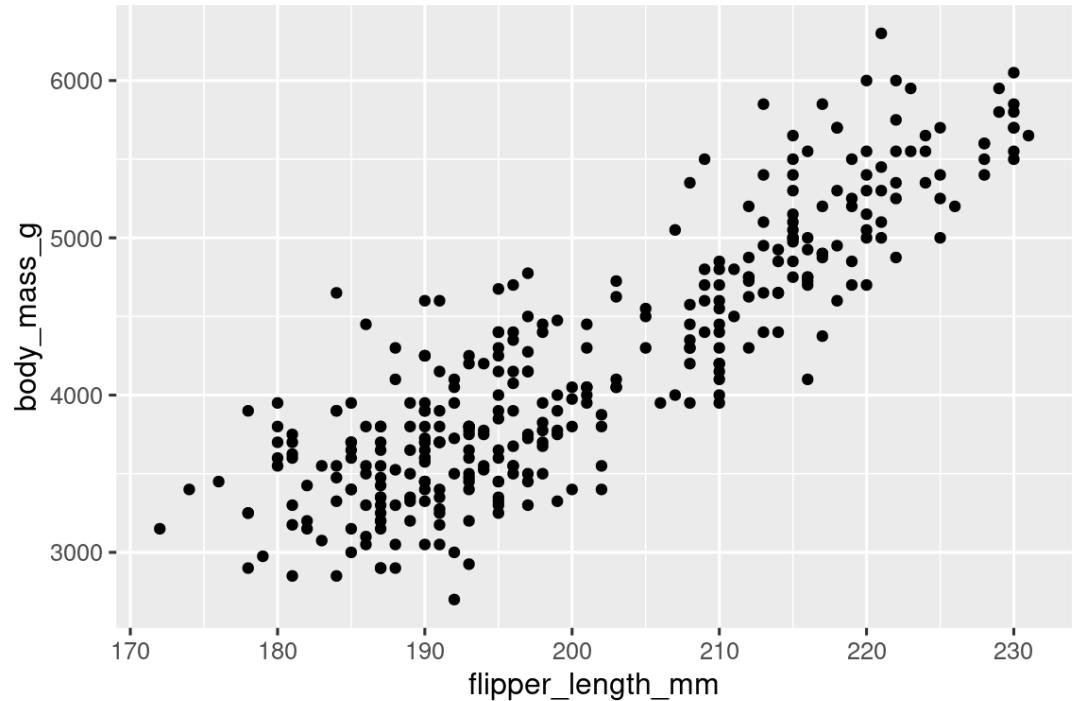


Data Visualization

Relationship – Two Quantitative Variables



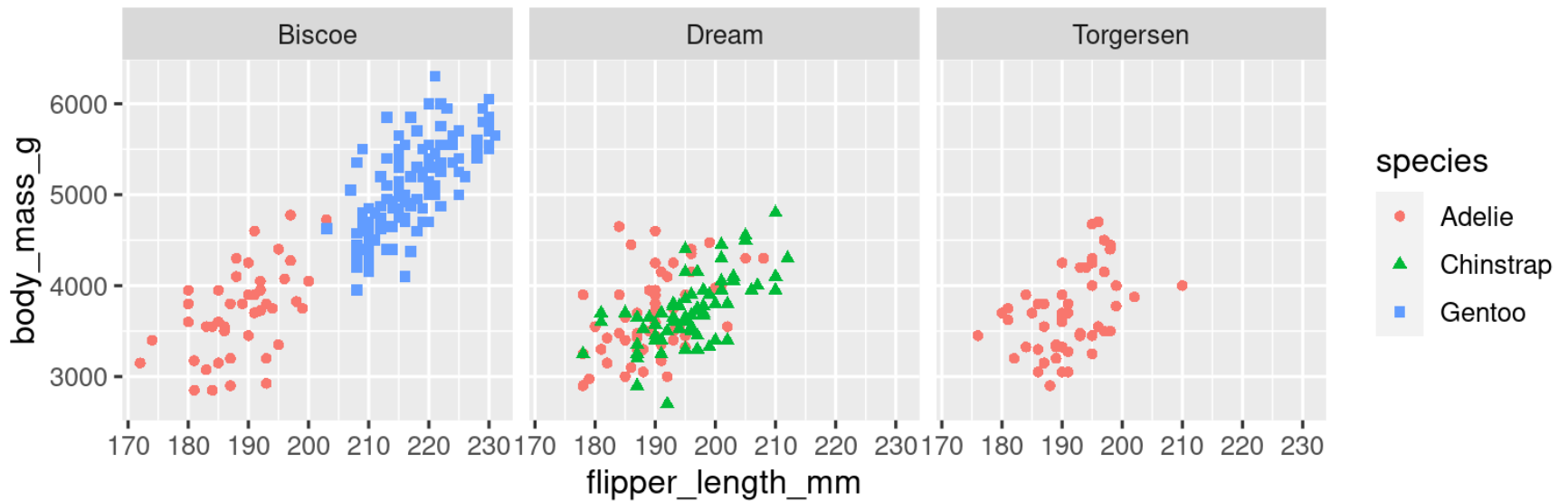
```
ggplot(penguins,  
aes(x =  
flipper_length_mm, y  
= body_mass_g)) +  
geom_point()
```



Data Visualization

Relationship – More
than Two Quantitative
Variables

```
ggplot(penguins, aes(x = flipper_length_mm, y  
= body_mass_g)) +  
  geom_point(aes(color = species, shape =  
species)) +  
  facet_wrap(~island)
```



Data Visualization

Saving your Plot

```
ggplot(penguins, aes(x = flipper_length_mm, y =  
body_mass_g)) + geom_point()  
ggsave(filename = "penguin-plot.png")
```